

Computing Leaves the Device

Ambient power, distributed intelligence, and the architecture of the spatial internet

1. The Last Generation to Look at Screens

I want to describe the world I actually believe we are heading toward, because every technical argument in this paper exists in service of it.

Start with the gramophone.

In 2005, a gramophone was not a broken object. It worked. It played music, faithfully, to anyone who wanted to hear music. What killed it was not a better gramophone. It was a format change that made its entire physical form pointless — the needle, the groove, the horn, the furniture-sized cabinet, all of it existed to solve a problem that had stopped existing.

Screens are gramophones.

A screen is a window you look into. Every screen ever manufactured is the same compromise: take a three-dimensional world, flatten it onto a plane, and ask a brain evolved for depth, parallax, and presence to accept the flattening. We have become extraordinarily good at that compromise. 4K, HDR, 120 hertz, quantum dot, OLED — decades of brilliant engineering spent making a lossy translation less noticeable.

The translation is about to stop being necessary.

Before describing what replaces it, here is the single thing this paper optimizes for, because every architectural decision in it follows from one objective:

Drive the power requirement of every device down far enough that it can be powered wirelessly.

That is the whole argument. Everything else — computation moving off the body, inference nodes in buildings, deciding what to transmit versus what to reconstruct, robots buffering their own power spikes — is a means to that one end. Each is a way of removing work from the device, and every unit of work removed is a unit of power the device no longer has to carry.

Get the demand low enough and something changes in kind rather than in degree: the device stops needing stored energy at all. It draws what it needs

from the space around it, continuously, the way it already draws Wi-Fi. No plugging in. No docking. No charging overnight.

That is the end state — a wirelessly powered world — and the rest of this paper is the architecture that gets there.

One clarification on scope, because it determines whether any of this is worth reading. I started by asking how to power better phones and laptops, and it was the wrong question. Those devices are the current form of a compromise that is ending. Designing a power architecture for them is designing for a category on its way out.

The devices worth designing for are the ones that come next: no screens, no batteries, sensing on the body, intelligence in the environment, worlds rendered around you rather than displayed in front of you. Both halves have to be built for that — what transmits the power, and what receives it. Solving for the devices we already have gets you a slightly better phone. Solving for the devices that replace them gets you the infrastructure everything after runs on.

So, what replaces the screen.

A recorded moment today becomes footage: a rectangle you watch, from the position the camera happened to occupy, framed by whatever the person holding it chose to include. In the world I am describing, you do not watch it. You stand inside it. You turn your head and the scene responds the way a real room responds. You move and the perspective changes the way perspective changes. It is not a recording of a place. It is the place.

Video calls stop being a grid of rectangles containing faces. You meet somewhere. A landscape, a workshop, a room that matches what the conversation is about — generated on demand rather than loaded from a library, rendered instantly, specific to that meeting. A colleague eight thousand kilometers away stands beside you at the machine you are both trying to fix, at the correct scale, casting light-consistent shadows, and points at the part they mean.

Presence is not a resolution problem. The brain has a threshold, and it is lower than the industry assumes. What convinces it is not pixel density. It is latency short enough to be imperceptible, parallax that responds correctly to your head, occlusion that behaves, sound arriving from the right direction with the right reflections, and lighting consistent with the geometry you are standing in. Get those aligned and the brain stops auditing the experience and simply accepts it. We have all felt that threshold cross in a good dream. It does not require perfect fidelity. It requires coherent fidelity.

And none of it plugs in.

The devices that deliver this draw milliwatts to a few watts — harvested

continuously from the field around them, supplied by localized emitters in buildings and, eventually, by solar collectors in orbit beaming power to the ground. Nothing charges overnight. Nothing docks. Nobody carries a cable, monitors a percentage, or plans a day around an outlet. The verb leaves the language the way “dialing” left it.

Here is the consequence that I think is most underrated.

The battery is the most tyrannical constraint in modern industrial design. Almost every consumer device of the last thirty years is, structurally, a box built around a cell. Thickness, weight, weight distribution, thermal layout, silhouette, materials, the entire visual language of consumer electronics — all downstream of a rectangular lithium brick and the heat it produces. Designers have spent a generation hiding it, and the best of them made hiding it look like art.

Remove it and industrial design gets its freedom back for the first time since the transistor.

Glasses that look like glasses, because they weigh what glasses weigh. Objects that compute and look like jewelry, like fabric, like nothing in particular. Forms determined by who the object is for and what it does, rather than by what has to fit inside it. Not five form factors defended by five companies, but a thousand, made by anyone, in materials nobody has been able to use because they could not dissipate heat or accommodate a cell.

That is a creative explosion, and it follows a pattern we have seen before. When a constraint that shaped an entire category lifts, the first decade is chaos and delight. It happened when synthesizers freed music from what a body could physically play. It happened when digital cameras freed photography from the cost per frame.

This is what abundance looks like at the scale of a Kardashev Type I civilization, and the Type I threshold is less exotic than it sounds. It does not mean starships. It means energy so available that it stops being a design consideration at all — the way clean water stopped being a design consideration in a modern city, the way bandwidth stopped being one for most of what we do online. Ambient energy across a room and orbital energy across a continent are the same idea, and both are steps along the same path.

I find this future overwhelmingly optimistic, and I also think it is closer than it looks, for a specific reason: not one piece of it requires a scientific miracle. Every component is an engineering problem, most of them already have people working on them for entirely unrelated reasons, and the funding is already flowing.

None of it exists yet. The reason is narrower and more specific than almost anyone assumes.

2. The Battery Is the Last Unsolved Systems Problem

Every component of modern computing has improved by orders of magnitude. One has not.

Since the early 1990s, compute per dollar has improved by something on the order of a million times. Storage cost per gigabyte has fallen by a factor of roughly a hundred thousand. Wireless throughput from 2G to 5G improved by five orders of magnitude. Image sensors, displays, radios, and machine learning models have all followed steep curves.

Lithium-ion energy density has roughly tripled or quadrupled in the same period. Commercial cells went from around 90 watt-hours per kilogram to something in the range of 250 to 300. That is an improvement of a few percent per year, compounding slowly, bounded by electrochemistry rather than lithography. There is no Moore's Law for electrons in a cell, because the constraint is not pattern resolution. It is the periodic table.

The consequence is that the battery has quietly become the organizing principle of hardware design. Look closely at any modern device and you find a system architected around energy scarcity rather than around computation:

- Phones ship with thermal throttling because sustained compute drains the cell faster than the cell can be replenished.
- Wearables run stripped-down processors not because engineers lack better silicon, but because better silicon would empty the battery before lunch.
- Laptops perform differently on battery than on mains power, which means the same machine is two different computers depending on where its electrons come from.
- Every major smartphone release allocates a growing share of its internal volume to the cell rather than to the computer.

We have normalized this. We treat "battery life" as a product specification rather than as evidence of an unsolved engineering problem. But the pattern is clear once you look for it: for thirty years, the industry has responded to the energy constraint by making everything else more efficient, and then spending that efficiency on more features until the constraint reasserts itself.

The interesting question is not how to make batteries better. Chemists have been working on that for decades with steady, incremental, honest results. The interesting question is different:

What would computing look like if we stopped designing around the battery entirely?

That question turns out to reorganize almost everything downstream of it.

3. Computing Evolves by Moving Bottlenecks

The history of computing is not a history of solved problems. It is a history of relocated ones.

Mainframes were bottlenecked by physical access. Solution: put the computer on the desk. Personal computers were bottlenecked by isolation. Solution: network them. Networked PCs were bottlenecked by local storage and local software maintenance. Solution: move both to the cloud. Cloud-dependent devices were bottlenecked by connectivity. Solution: mobile broadband. Mobile devices were bottlenecked by their own processing limits. Solution: server-side inference.

At no point did anyone eliminate a constraint. They moved computation to wherever the constraint was cheapest to satisfy.

Streaming is the cleanest example. Music did not solve local storage. It relocated storage to a data center and paid for that relocation in bandwidth, which was falling in cost faster than storage was growing in capacity. The iPod's entire value proposition — carry your library with you — was dissolved not by a better iPod but by an architectural decision that made the question irrelevant.

This is the pattern worth internalizing, because it tells you what the next move looks like. Each generation asks: *which resource is getting cheaper fastest, and what can I move onto it?*

Right now, two resources are getting dramatically cheaper: distributed computation and wireless bandwidth. One resource is nearly static: portable stored energy.

The architectural implication is not subtle. **If energy storage is the slow curve, move work off the things that carry stored energy.** Not by making devices more efficient at the margin, but by relocating the energy-hungry functions to places where energy is effectively unlimited.

The next migration in computing is not a faster processor. It is the migration of computation and energy away from the body.

4. Compute Does Not Belong on the Human Body

Wearables inherited their architecture by accident.

When smartwatches and headsets were designed, the only reference architecture available was the smartphone: a self-contained unit that senses, computes, renders, communicates, and stores its own power. That model was correct for phones. It was copied onto the body without anyone asking whether it should be.

The result is a category of devices asked to do five jobs simultaneously, in a thermal envelope constrained by human skin, powered by a cell constrained by human tolerance for weight. A modern mixed reality headset performs simultaneous localization and mapping, hand and eye tracking, scene understanding, neural rendering, and high-bandwidth radio, all while remaining light enough to wear and cool enough to sit against a face. Every one of those requirements fights the others.

This architecture was already strained. As models grow larger and more capable, it becomes untenable. The compute demanded by useful spatial intelligence is rising faster than the energy budget available on a human head, and there is no version of that curve where the head wins.

The way out is to stop treating the wearable as a computer.

The device on your body should be a sensor and a display. Nothing else.

Its job is to capture what the world looks like from your position, transmit that efficiently, receive a rendered or partially rendered result, and put it in front of your eyes. Cameras, IMUs, microphones, eye trackers, and a radio. All of those can run in the milliwatt range with existing silicon. What cannot run in the milliwatt range is inference and rendering — so those should not be there.

Glasses that weigh thirty grams and last all day are not a distant materials science achievement. They are an architectural decision that has not yet been made at scale.

5. The Shift from Geometry to Intent

If rendering moves off the device, something has to travel over the air. The question of what travels is the central technical question of the spatial internet, and both obvious answers are wrong.

Answer one: send the geometry. Capture the world volumetrically, compress it, stream it. This is the direct descendant of video streaming, and it has the virtue of being conceptually simple. It also has the vice of enormous bandwidth cost. Raw volumetric and light-field capture runs into gigabits per second before compression. You can push that down substantially with good codecs, but you are fighting a losing battle: fidelity scales with data, and

human vision is demanding.

Answer two: send the meaning. Transmit a compact semantic description — this is a room, there is a person here, they are moving this way — and let a generative model reconstruct the scene at the receiving end. Bandwidth collapses to almost nothing. But you have transferred the entire cost to inference, which means power, heat, latency, and a reconstruction that is plausible rather than accurate. For entertainment that may be fine. For a surgeon, an engineer, or anyone whose decisions depend on the world being exactly as it is, it is not.

Both answers optimize the wrong variable. One minimizes computation. The other minimizes transmission. Neither minimizes what actually constrains the system.

The correct framing:

Transmit only the information whose transmission costs less energy than reconstructing it locally.

That reframing turns a format debate into an optimization problem. Geometry, semantic descriptors, latent features, neural scene representations, raw sensor measurements, and cached priors are not competing standards. They are interchangeable tools inside one energy budget. A static wall that has not changed in an hour should not be transmitted; it should be recalled. A person's face mid-expression should be transmitted with high fidelity, because generating it is both more expensive and less trustworthy. A texture the receiver has seen a thousand times should be a pointer, not a payload.

The objective function is not bits. It is total energy consumed across communication, computation, memory, and rendering, subject to perceptual fidelity and real-time responsiveness.

This matters because the trade-off does not stabilize. 6G will relax bandwidth constraints without eliminating them. Better inference silicon will lower computation costs without zeroing them. The ratio between “cheaper to send” and “cheaper to generate” will keep shifting — by generation, by location, by time of day, by whether you are in a stadium or a forest.

Which means the architecture cannot pick a side. **It has to negotiate the split dynamically, per scene, per frame, per object.** That negotiation layer is the piece nobody has built yet, and it is more important than any individual codec or model.

6. The Rise of Distributed Intelligence

If the device stops generating worlds, the generation has to happen somewhere. There are two default answers and both fail.

The cloud fails on physics. Comfortable spatial computing requires motion-to-photon latency under roughly twenty milliseconds. Light in fiber travels around 200,000 kilometers per second, so a round trip of a thousand kilometers costs about ten milliseconds before a single packet is queued, processed, or rendered. Hyperscale data centers are built where land and power are cheap, which is almost by definition far from where people are. The speed of light is not an engineering problem to be optimized. It is a boundary condition.

The local device fails on energy. That was section four.

What sits between them is not a compromise. It is a new layer: **localized micro data centers positioned within a few kilometers of users.** Cabinets, not campuses. Enough compute to render for a neighborhood, close enough that the round trip is a millisecond or two, connected to mains power and to real cooling, and — this is the part that matters — shared across everyone in range.

Sharing is what makes the economics work. A headset's GPU is idle most of the time and sized for peak load. A neighborhood inference node runs at high utilization because human activity is statistically smooth in aggregate. The same silicon serves a hundred users at a fraction of the per-user cost, in a location where cooling is a fan rather than a thermal-comfort constraint against someone's forehead.

This is the same logic that moved computation from desks to data centers, applied one step back in the other direction. Not centralization. Not decentralization. Placement.

Once that layer exists, the hierarchy inverts. The node becomes the computer. The glasses, the watch, the phone, the car, the robot — all become interfaces to it.

7. Spatial Computing Becomes Infrastructure

The moment computation becomes room-scale rather than body-scale, it stops being a consumer electronics problem and becomes a **building** problem.

This transition has happened before, repeatedly, and it always looks the same. Electricity began as a per-device consideration and became a wall socket. Networking began as a cable you ran yourself and became a property feature.

Air conditioning began as a machine you bought and became a system you inhabit. In each case, the technology stopped being a product you carried and became a service the space provided.

Compute is next.

An office already has an electrical plan, a network plan, and an HVAC plan. It will acquire a compute plan: where the inference nodes live, how they are cooled, how much power they draw, how coverage is distributed so that no room is a dead zone. A hospital will specify computational coverage in an operating theater the way it currently specifies gas lines and sterile airflow. A factory will treat spatial inference as infrastructure alongside compressed air.

Real estate developers will advertise it. Building codes will eventually address it. Landlords will meter it.

And the room, having acquired sensors and inference and memory of its own state, becomes computationally intelligent — not in a mystical sense, but in the ordinary sense that it can perceive its own geometry and occupancy and serve that understanding to whatever devices are present.

The wearable stops being a computer you wear and becomes a key you carry.

8. The New Power Hierarchy

Now the argument closes on itself, because relocating computation relocates energy demand.

The principle is simple: **power should exist where computation exists.**

- **High-power computation belongs on mains electricity.** Inference nodes, rendering, model execution, anything measured in watts or hundreds of watts. These live in buildings, plugged into a grid that already delivers effectively unlimited energy to fixed locations.
- **Medium-power systems belong on wheels, rails, and rechargeable platforms.** Vehicles, robots with docking stations, equipment that returns to a charger as part of its normal duty cycle.
- **Low-power sensing belongs on the body.** Cameras, IMUs, microphones, radios, displays. Milliwatts, and with aggressive duty cycling, microwatts on average.

Read that hierarchy against the history of wearable design and the mistake becomes obvious. We have spent a decade trying to fit watt-scale computation into a milliwatt-scale power envelope, and calling the resulting compromise a product.

The alternative is not a better battery. It is a smaller problem.

Instead of making batteries larger, minimize what requires a battery at all.

If the body-worn device draws single-digit milliwatts averaged over a day, the entire energy conversation changes character. You are no longer asking how to store enough energy for eight hours of GPU work. You are asking how to supply a few milliwatts continuously to something that never has to be plugged in. Those are not the same question, and the second one has answers the first one does not.

Energy follows computation. Move the computation, and the energy problem moves with it — into buildings, where it was already solved.

9. Ambient Power Instead of Managed Power

Here is where wireless power stops being a gimmick.

Wireless power today is mostly point-to-point: a charging pad with a coil, or a beam-steering system that tracks a device and delivers watts to it directly. Both are attempts to replace the cord. Both inherit the cord's mental model — a dedicated link between a source and a sink, negotiated, exclusive, and efficiency-obsessed, because when you are moving watts, every percentage point of loss is heat and cost.

That entire framing dissolves once device demand collapses into the milliwatt range.

At milliwatt scale, the objective is no longer efficient beam delivery. It is **coverage**. Energy stops being a link and becomes a field: continuously present, broadly distributed, harvested opportunistically by any device inside it, shared simultaneously by every device in range without negotiation or handshake.

The right analogy is not the power cord. It is radio.

Nobody asks whether an FM signal is efficiently delivered to their car. The transmitter radiates, and anyone within coverage receives. The system is enormously wasteful by point-to-point standards and completely successful by its own, because the value is in ubiquity rather than in efficiency.

Ambient power works the same way. A transmitter saturating a volume with RF at safe exposure limits delivers a low but reliable power density everywhere inside it. Any device with an antenna and a rectifier harvests what it needs. Ten devices in the room do not divide a budget; they each take what they can capture. Adding a device costs the infrastructure nothing.

Honest accounting matters here, because this is where the field loses

credibility. Ambient RF harvesting from existing broadcast and cellular signals yields microwatts at best — enough for a sensor that wakes once a minute, not enough for continuous operation. Purpose-built ambient power systems can do considerably better within safety limits, but the delivered power density still falls off with distance and still has to respect exposure standards that exist for good reasons. The end-to-end efficiency of such a system, measured conventionally, is poor.

The response to that is not to dispute the numbers. It is to reject the metric.

Efficiency is the wrong measure for infrastructure whose value is coverage. The grid loses a meaningful fraction of generated power in transmission and distribution, and no one considers this an argument against electrification. Broadcast radio delivers a vanishing fraction of radiated power into any given receiver. The question is not what fraction arrives. The question is whether enough arrives, everywhere, for the thing you want to run — and whether the total energy consumed is small enough that the waste does not matter.

A city-scale ambient power layer supplying billions of milliwatt devices consumes a rounding error against the same city's air conditioning load. That is the correct comparison, and it favors ambient power decisively.

10. The Four Converging Curves

No single breakthrough produces this architecture. It emerges from four independent trends, none of which were created for this purpose.

Curve one: falling device power. Sensor front ends, low-power radios, always-on wake circuits, and near-threshold silicon have driven the standby and duty-cycled power of a capable sensing device down by orders of magnitude over two decades. This curve is driven by the IoT and mobile industries and continues regardless of anything in this paper.

Curve two: improving harvesting efficiency. RF-to-DC rectifier design, low-voltage cold-start circuits, and multi-band antenna design have steadily improved the fraction of a weak field that can be converted into usable power at low input levels. This is a niche field advancing on its own timeline, driven largely by RFID, medical implants, and industrial sensing.

Curve three: compute decentralization. Edge inference silicon, telecom edge deployments, and the economics of latency-sensitive services are pushing computation outward from hyperscale centers toward users. This is being funded by the entire telecommunications and cloud industry for reasons entirely unrelated to wearables.

Curve four: falling launch costs. Reusable launch has cut the cost of putting mass into orbit by roughly an order of magnitude within a generation, with further reductions plausible. This curve is driven by satellite constellations and national space programs.

Each curve is independently financed, independently motivated, and independently on track.

The systems argument is that they multiply rather than add. Device power falling by three orders of magnitude and harvesting efficiency improving by one means the gap between what a device needs and what a field can supply closes by four orders — and that gap closing is the entire precondition for battery-free operation. Compute decentralization is what allows device power to fall in the first place, because the work has somewhere to go. Falling launch costs extend the same architecture beyond terrestrial infrastructure.

Nobody is building toward this convergence. It is happening as a byproduct, which is the most reliable kind of technological progress there is.

11. Why AI Accidentally Solves Wireless Power

The AI infrastructure buildout looks like the opposite of a low-power future. Gigawatt data centers, dedicated power plants, and a capital expenditure race measured in hundreds of billions of dollars. It reads as a story about energy consumption, not energy liberation.

It is, in fact, the single strongest enabling force for battery-free devices, and the mechanism is almost too simple to notice.

Every dollar spent making remote inference cheap and fast reduces the amount of computation that must happen locally.

The industry is building, at enormous scale and for entirely commercial reasons, the exact infrastructure this architecture requires: dense inference capacity, low-latency distribution, model serving at scale, and the software stack to route work to it. The buildout is currently oriented toward text and images. The physical assets are indifferent to what they render.

Nobody in that industry is trying to eliminate batteries. They are trying to serve models profitably. But the second-order effect of succeeding is that the computational floor for a client device keeps dropping — and the lower that floor falls, the closer it gets to what an ambient field can supply without any battery at all.

This is how infrastructure transitions usually work. The enabling investment is made for an unrelated reason, by people with no interest in the eventual application, and the application arrives later to find its foundation already

poured.

12. The Spatial Internet

Put the pieces together and the result is not a better headset. It is a different internet.

The internet we have is visual and symbolic. Information lives inside rectangles. We navigate it through metaphors — pages, files, folders, windows, feeds — that were invented to make an abstract system legible on a two-dimensional screen. Every interaction requires translating an intention into symbol manipulation, and every device is a portal you look *into*.

The spatial internet inverts that. Persistent three-dimensional environments, continuously generated around you, anchored to real locations, shared with other people who are physically or virtually present. Information does not live in a rectangle. It lives where it is relevant: the maintenance history hovering beside the machine, the annotation attached to the physical wall, the colleague rendered in the chair across from you.

Three shifts follow:

- **The internet becomes spatial rather than visual.** Content has position, persistence, and physical relationship rather than URL and page rank.
- **Interaction becomes physical rather than symbolic.** You reach, move, look, and speak. The translation layer between intention and action thins.
- **Interfaces disappear into environments.** The most successful interface is the one nobody notices, because the room simply behaves correctly.

This is only possible under the architecture described above. A spatial internet running on self-contained battery-powered devices is a demo. A spatial internet running on ambient power and distributed inference is infrastructure.

13. Robotics Changes the Equation Again

Robots are currently designed exactly the way wearables are: as complete self-contained computers that carry their own perception, planning, inference, and power.

That design faces the same constraint, amplified. A humanoid robot spends a substantial fraction of its energy budget carrying the mass of its own battery and computer, and a further fraction cooling that computer. Every watt of onboard inference is a watt not available for actuation, and every kilogram of

battery is a kilogram of payload lost.

Once ambient sensing, distributed inference, and spatial intelligence mature, robots inherit the architecture for the same reason wearables do. A robot operating inside a computationally instrumented building does not need to build a map — the building has one. It does not need to run full scene understanding — the room is already doing it, for everyone. It becomes an **embodied endpoint**: actuators, sensors, safety-critical local reflexes, and a fast link to the infrastructure that handles everything above the reflex layer.

Local autonomy does not disappear. Anything safety-critical stays on board, because a robot that stops thinking when the network hiccups is not a product. But the split between what must be local and what should be infrastructural is an engineering decision, and current designs make it in the wrong place by default.

The battery backpack is the wrong answer

The industry's response to robot endurance has been to pursue better chemistry — solid-state, lithium-metal, anything that raises energy density. The goal is a better battery backpack.

This treats the symptom. A physical agent that requires tens of kilograms of stored energy to operate for a few hours has already lost the architecture argument, and it loses it twice: once because it spends a meaningful share of its energy carrying and cooling its own power supply, and again because that mass degrades every mechanical parameter that matters.

It also forecloses the endgame. A robot with a high continuous power draw cannot be powered wirelessly, and the reason is not engineering difficulty. It is safety. Delivering kilowatts continuously to a moving target in a public space requires field intensities far above human exposure limits. At those densities you have not built power infrastructure. You have built a directed-energy weapon and pointed it down a sidewalk.

Exposure limits are not a regulatory inconvenience to be lobbied away. They are the hard ceiling the entire architecture has to fit under.

Trickle-spike

Once you accept the ceiling as fixed, the design problem inverts. Stop asking how much power can be transmitted. Start with what can safely be harvested, and build the machine that fits inside it.

The baseline available to a wirelessly powered robot is governed by a first-principles relationship:

Harvested power = incident power density × effective receiving area × conversion efficiency

Three terms, and only one of them is locked.

Power density is capped by safety. It sits wherever exposure standards put it, and no amount of transmitter engineering moves it. Treat it as a constant.

Effective area is a design variable, and a larger one than it appears. The naive reading is that a robot needs to carry a flat panel — which is both mechanically absurd and aesthetically disqualifying. Conformal metamaterial rectennas remove that constraint. The receiving surface becomes the skin of the machine: curved, distributed across torso and limbs, integrated into structure rather than bolted onto it. The collecting area is the robot's own surface, not a plate it has to hold up.

This term is honestly the messiest one, and the paper should say so. Effective area is not fixed — it is the projected area presented to the incident field, and it changes continuously as the robot turns, bends, reaches, and walks. A machine facing the source collects differently than one turned away. Real systems will need multi-directional transmitters, distributed receiving surfaces that keep some portion illuminated in any pose, and a design discipline that treats orientation-averaged collection as the number that matters rather than best-case collection. Anyone quoting a single fixed area for a moving machine is quoting a best case.

Conversion efficiency is an engineering variable — rectifier design, impedance matching across incident angles, low-voltage behavior — and it is improving for reasons unrelated to robotics.

Set against that harvestable baseline is the demand side, which is where the real work happens. Two levers, and the specific technology behind each matters less than the direction.

Compute that costs less. Neuromorphic silicon, event-driven perception, photonic approaches, aggressive offload of everything above the reflex layer to surrounding infrastructure — these compete with each other, and the competition is beside the point. Any of them that moves onboard computational draw down by orders of magnitude rather than percentages serves the architecture equally well.

Actuation that fights physics less. Passive dynamics, elastic energy storage, compliant mechanisms, bio-inspired designs that exploit gravity and momentum instead of brute-forcing every joint with oversized motors. Biological locomotion is efficient because it spends most of its cycle falling and catching. Whichever approach gets closest to that, wins — and the

architecture is indifferent to which one does.

The part that makes it work

Even with demand driven down, robots have a load profile problem: locomotion is not a flat draw. Standing, sensing, and slow movement sit near the baseline. Jumping, lifting, catching, and sprinting spike an order of magnitude above it, for fractions of a second.

Sizing a power system for those spikes is what produces the battery backpack. But the spikes do not need to be *supplied* by the harvest. They need to be *buffered*.

That is the trickle-spike architecture:

Harvest the baseline continuously and wirelessly. Buffer the peaks locally in onboard storage.

Solid-state supercapacitors are well suited to exactly this role — high power density, fast discharge, high cycle life, and indifference to the deep-discharge behavior that degrades chemical cells. Paired with a small battery for reserve, they absorb the spike and recharge from the trickle during the far longer intervals between spikes.

The energy balance is straightforward: over any duty cycle, harvested energy must meet or exceed consumed energy. Storage decouples instantaneous supply from instantaneous demand, which means the machine can momentarily draw far more than the field delivers, provided the average closes.

This is the reframing that matters. **Onboard storage stops being sized for endurance and becomes sized for the largest single burst.** Endurance comes from the field. Storage handles transients. Those are different components, orders of magnitude apart in capacity, and conflating them is what put a battery backpack on every humanoid in development.

The claim here is not a specific number. It is that this equation is the correct frame, and that every term in it except the safety ceiling is a live engineering variable moving in a favorable direction. As long as the arithmetic closes — orientation-averaged collection at safe densities, against a demand curve flattened by whichever compute and actuation approaches win — ambient-powered robotics is a systems design problem rather than a physics problem.

That distinction is the entire argument.

The physical world becomes the operating system, and robots become one of several classes of process running on it.

14. Beyond Earth: Orbital Energy Infrastructure

Space-based solar power has been proposed since the 1960s and has failed on economics every time. The physics was always sound: a collector in orbit receives sunlight continuously, without atmosphere, weather, or night, at roughly a third more intensity than the surface. The problem was that lifting a power station cost more than the power was worth by a margin that no efficiency improvement could close.

Launch cost is the only variable that ever mattered, and it is the one that moved. Reusable launch has already cut cost-per-kilogram to orbit by an order of magnitude, and further reductions are credible if not yet demonstrated.

Meanwhile, the transmission side has moved from theory to hardware. Orbital demonstrations have beamed measurable power to receivers on the ground. The quantities are small — this is early-stage work, and honest reporting means saying so — but the loop has been closed, and closing a loop is the hard part.

The framing that matters: **orbital power is not a separate technology. It is the same ambient architecture at a different altitude.**

A terrestrial ambient power layer and an orbital one solve the same problem — delivering energy to distributed receivers without physical connection — with the same fundamental method. The receiver technology overlaps. The regulatory questions rhyme. The system design principles are identical. One serves a room; one serves a region.

This is why orbital power belongs in this argument rather than in a separate speculative appendix. It is the outermost layer of the same stack, and it becomes plausible for a reason unrelated to itself: launch economics improved because of satellite internet, and orbital power gets the benefit.

15. A New Computing Stack

The traditional stack is device-centric and terminates in the constraint we started with:

```
Application
  ↓
Operating System
  ↓
Hardware
  ↓
Battery
```

Everything above the battery is negotiating with it. Every layer's design is partly a concession to the layer beneath, and the bottom layer improves at a few percent a year.

The replacement is environment-centric:



The distinction is not cosmetic. In the first stack, the unit of computing is a device, and the device is bounded by what it can carry. In the second, the unit of computing is an environment, and the environment is bounded by what the grid can deliver — which is to say, not meaningfully bounded at all.

Note what happened to the bottom layer. It did not get better. It got *replaced by infrastructure*. That is the same move electrification made when it replaced on-site generators with a grid, and the same move cloud computing made when it replaced server closets with regions. In both cases the winning architecture was not a superior version of the constrained component. It was the removal of the component from the critical path.

16. Capital Always Builds the Future Before It Understands It

Infrastructure consistently arrives before the application that justifies it, built by people who are wrong about what they are building.

Railroads were financed for freight and reorganized where people lived. The electrical grid was built to sell lighting and enabled the assembly line, refrigeration, and everything downstream of the electric motor. The fiber laid during the telecom bubble was justified by demand forecasts that were badly wrong, went dark, sold for cents on the dollar, and then carried streaming video and the cloud. Data centers built for enterprise virtualization became the substrate for everything that followed. The AI buildout underway now is justified by revenue projections that will look either naive or timid in a decade, and either way the concrete and copper will remain.

The pattern: capital builds capacity for a stated reason, the stated reason turns out to be incomplete, and the capacity gets used for something its

financiers did not anticipate.

Every component of the architecture in this paper is already being financed by someone with an unrelated motive. Edge compute is being built for latency-sensitive commercial services. Low-power silicon is being built for connected devices. Harvesting circuits are being refined for industrial sensing and implants. Launch capacity is being built for satellite constellations. Inference capacity is being built for a market that has nothing to do with spatial computing.

Nobody is funding “ambient-powered computing,” and nobody needs to. **The future described here emerges from the intersection of investments already being made, not from a deliberate program to create it.** That is the strongest reason to believe it happens, and also the reason it will arrive without an announcement.

17. What Can Be Built Today

A vision that requires everyone to wait is not a vision. It is a forecast, and forecasts do not get funded.

So here is the practical question: what can be built and sold profitably in the next three years that also moves this architecture forward? Every item below passes one test.

It has a paying customer who does not care about this paper.

That constraint is not a compromise. It is the mechanism. Infrastructure gets built when each layer is independently viable, and the businesses that end up owning a transition are almost always the ones that were solving a boring problem for real money while the transition happened around them.

Shipping now: businesses with customers today

Ultra-low-power perception silicon. Sensors and front ends that detect, classify, and wake at microwatt averages — event-driven cameras, always-on audio front ends, near-threshold DSPs. The buyers exist already: industrial condition monitoring, automotive occupant sensing, hearables, security. Nobody in that market is thinking about spatial computing. Every design win drives curve one, and the resulting silicon is exactly what a battery-free wearable needs.

Battery-free sensing for buildings and logistics. The clearest wedge for ambient power is not consumer devices. It is the coin cell. Commercial

buildings, cold chain, warehouses, and retail shelf systems deploy sensors by the thousand, and the dominant lifetime cost is a technician walking around replacing batteries. Harvested-power sensors that never need that visit sell on maintenance economics alone, in an unglamorous market with real budgets. This is where harvesting circuits, low-voltage cold start, and deployment experience get paid for.

Split-compute wearables for enterprise. Glasses that offload rendering and inference to a nearby node, sold into field service, warehousing, surgical support, and inspection — environments where the user is stationary or in a known building, where a local node can be installed as part of the deployment, and where the customer measures value in error rates rather than in delight. Enterprise tolerates a tether that consumers will not. It is the fastest legitimate path to proving the offload architecture at scale.

Latency-critical edge inference nodes. Micro data centers placed near population density, sold to whoever needs single-digit millisecond round trips: cloud gaming, live production, industrial vision, autonomous systems, and increasingly ordinary AI inference where round-trip time is the product. The economics resemble real estate and telecom more than software — capital-intensive, debt-financeable, defensible by location. Whoever builds this footprint for gaming and inference owns the substrate spatial computing later runs on.

The strategic middle: highest leverage, least crowded

The adaptive representation layer. This is the most valuable unbuilt thing in the entire architecture and it is a software business. A system that decides, continuously, what to transmit and what to reconstruct — geometry, features, semantics, or a cached pointer — based on available bandwidth, latency budget, and energy cost on both ends.

It has customers now, before any of the rest of this exists: remote robot and drone teleoperation over unreliable links, cloud rendering, volumetric telepresence, and defense communications in contested spectrum. All of them are currently solving it badly and privately, inside their own stacks.

Build it well, publish the benchmarks, and it becomes the standard by default. Codecs are how you own a medium without owning any of the hardware.

Persistent spatial scene services. The reusable prior — a maintained, queryable, up-to-date representation of a real building — sold today to construction, facilities management, insurance, and robotics integrators. Everyone wants the scan. Almost nobody has solved keeping it current or serving it in real time. That capability is a product now and a dependency later.

Infrastructure-side robot autonomy. Sell warehouse and factory operators

a reduction in cost per robot by moving perception and planning off the robot and into the building. The pitch is procurement arithmetic: cheaper units, longer runtime, fleet-wide upgrades in software. The result is the first commercially instrumented environments, which is the same infrastructure everything else in this paper needs.

Compute as a building system. Specification, integration, and operation of inference capacity as part of a building's design — power, cooling, coverage, placement — sold to developers and large operators the way structured cabling and building management systems are sold. Low-margin, unfashionable, and the reason it happens on schedule.

Capital-intensive: fundable now, returns later

Power beaming for places wiring cannot reach. Remote installations, off-grid sensing, drone endurance, and defense logistics have budgets today and no good alternative. The receiver architecture, phased-array transmission, safety validation, and regulatory groundwork developed for those customers transfer directly to both terrestrial ambient power and orbital delivery.

Orbital power components, not orbital power stations. The station is not the investable object yet. The components are: lightweight high-efficiency photovoltaics, thermal management, phased-array transmitters, deployable structures, and ground rectennas. Every one has a satellite, defense, or terrestrial customer today, and space agencies across several countries are actively funding demonstrations. Build the pieces on someone else's dime and the station becomes an integration problem when launch costs finish falling.

What not to build

Three traps, each of which has consumed serious money already:

A fully self-contained consumer headset. It is the gramophone with a better needle. The constraint is architectural, and no amount of silicon or industrial design solves it from inside the device.

A battery breakthrough as the load-bearing assumption. Improve cells if you can, and take the incremental wins. But an architecture that only works if energy density triples is a bet against thirty years of evidence.

An ambient power company with no anchor customer. The technology is real and the timeline is long. Without a boring revenue-generating deployment paying the bills, the company dies in the gap between demonstration and infrastructure. That gap has killed nearly everyone who has tried.

For researchers

The commons matter as much as the companies here, and they are cheaper to create. The highest-leverage contributions are shared benchmarks for energy-per-perceptual-unit across representation strategies, open datasets for real spatial capture under real bandwidth constraints, honest published measurements of harvested power density in occupied environments, and reference implementations that let separate teams compare against each other instead of against their own last quarter.

Standards written early by people who understand the physics are worth more than any single product built on top of them.

18. Research Agenda

The previous section covered what can be sold. This one covers what nobody can sell yet, because the answers do not exist. These are the questions that decide how fast the rest arrives, and they are worth someone's career:

On device power - How low can the power draw of a fully capable sensing-and-display wearable realistically fall, given optimal duty cycling and an offloaded compute path? - What is the true floor for always-on perception — the wake circuits, the low-rate sensors, the radio that must be listening?

On representation - What representation minimizes total system energy for spatial communication, and how does the optimum shift with available bandwidth? - How do you build a negotiation layer that decides, per object and per frame, whether to transmit or reconstruct? - What is the perceptual cost function that should govern that decision, and how do you measure it honestly?

On placement - Where should micro inference nodes be positioned to minimize latency and cost across a real population density? - What is the economic crossover point between local and distributed inference, and which variables move it? - How should buildings be architected for pervasive computation — power, cooling, coverage, and the physical placement of nodes?

On energy - What is the minimum viable ambient power density for a useful class of devices, and what does it cost to deliver at room, building, and city scale? - How does end-to-end system energy compare against the battery-centric baseline when you account for manufacturing, replacement, and disposal? - What regulatory framework governs diffuse wireless energy — exposure limits, spectrum allocation, interference, and liability?

On embodied systems - What is the orientation-averaged effective collecting area of a conformal rectenna surface across a realistic locomotion cycle, measured rather than modeled? - How should transmitters be distributed so

that some usable portion of a moving machine stays illuminated in any pose? - What is the true floor for onboard compute in a robot with infrastructural offload, once safety-critical reflexes are accounted for? - How large must burst storage be, expressed as a function of peak actuation demand rather than operating duration? - Where does the energy balance close first — which class of machine, in which environment, becomes wirelessly sustainable soonest?

On the outer layer - How should orbital and terrestrial power infrastructure coexist technically and economically? - What receiver architecture serves both?

None of these have settled answers. Several are dissertations. A few are companies.

19. Conclusion — Computing Leaves the Device

This paper does not predict a gadget. It describes an architecture, and architectures outlive the products built on them.

The history of computing has been a history of moving work to wherever it is cheapest to perform. Mainframe to desk, desk to pocket, pocket to cloud, cloud to edge. Each move looked like a product decision and was actually a placement decision, and each one was obvious in retrospect and contested at the time.

The next move is to stop asking *how powerful can the device become* and start asking a different question:

Which parts of computing belong on the body, in the room, in the city, and eventually in orbit?

That question is the whole argument. Once it replaces the old one, a set of independently developing trends stops looking coincidental. Falling device power, improving harvesting, decentralizing compute, and cheapening launch are not four unrelated stories. They are four sides of one transition, and they are converging whether or not anyone plans it.

Which returns us to where this paper started.

Standing inside a recording instead of watching it, meeting someone in a place that did not exist an hour ago, wearing something that never charges — none of that is a hardware wish. It is what falls out of answering the placement question correctly. Every element of it is downstream of moving computation off the body and energy off the device. Get the architecture right and the experience is a consequence rather than a goal.

The Kardashev Type I threshold sits roughly four orders of magnitude above our current planetary energy use. We do not close that gap by improving batteries. We close it by learning to treat energy the way we already treat bandwidth and clean water: as ambient, distributed, and universally available rather than as a resource each device hoards for itself.

The device becomes lighter. The environment becomes smarter. Computation dissolves into the spaces we inhabit, and the battery — the last unsolved systems problem — stops being a problem by ceasing to be a requirement.

That is not a prediction about hardware. It is a question about placement, and the answer is already being built by people who think they are building something else.